

COEFFICIENTWISE LOG-CONCAVITY OF THE CUBIC MULTIPLE-POCHHAMMER SERIES

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ABSTRACT. Let $(f_n)_{n \geq 1}$ be a nonnegative log-concave sequence with no internal zeros, and define the cubic multiple-Pochhammer series $F_f(\mu; x) = \sum_{n \geq 1} f_n(\mu)_{3n} x^n / (3n-1)!$. Karp and Zhang established coefficientwise log-concavity of the analogous series for multiplicity $r = 2$ and gave numerical counterexamples for $r = 4$, leaving the cubic case open as their Conjecture 3.8. We prove it: for all $\mu, \alpha, \beta \geq 0$, every coefficient of the generalized Turánian $F_f(\mu + \alpha; x)F_f(\mu + \beta; x) - F_f(\mu; x)F_f(\mu + \alpha + \beta; x)$ is nonnegative. The proof is elementary and noncomputational. The degree- m coefficient is represented as a beta-binomial expectation, reducing the theorem to the monotonicity of a weighted residue-class binomial kernel; this monotonicity is established through a Bernstein-basis positivity certificate obtained from a root-of-unity evaluation. A coefficientwise strictness criterion and ordinary log-concavity in μ follow.

1. INTRODUCTION

A formal power series depending on a parameter $\mu \geq 0$ is called *coefficientwise log-concave* if, for every $\mu, \alpha, \beta \geq 0$, all coefficients of the generalized Turánian

$$F(\mu + \alpha; x)F(\mu + \beta; x) - F(\mu; x)F(\mu + \alpha + \beta; x)$$

are nonnegative. This property is stronger than ordinary log-concavity of $\mu \mapsto F(\mu; x)$ whenever the series converges at a positive value of x . It has been studied systematically for reciprocal-gamma, gamma-ratio, hypergeometric, and Fox–Wright series; see [4, 5, 6, 7]. Turán-type inequalities for hypergeometric and confluent hypergeometric functions have also been studied extensively [1, 2].

Karp and Zhang considered, for an integer $r \geq 2$, the modified multiple-Pochhammer series

$$(1.1) \quad F_{f,r}(\mu; x) = \sum_{n \geq 1} f_n \frac{(\mu)_{rn}}{(rn-1)!} x^n,$$

where (f_n) is nonnegative and log-concave. Karp and Zhang report that A. Ninh suggested this construction to them in a private communication in 2017 [7]; related published work on log-concavity and Turán-type inequalities appears in [9]. They proved coefficientwise log-concavity for $r = 2$, gave numerical counterexamples for $r = 4$, and left the cubic case $r = 3$ open [7, Conjecture 3.8]. This note proves that conjecture.

Karp and Zhang index the ambient sequence from $n = 0$, although the modified series itself begins at $n = 1$. The log-concavity hypothesis is stated here directly for the positive-index tail $(f_n)_{n \geq 1}$. This is equivalent to their formulation: restriction preserves log-concavity and interval support, while any admissible tail extends to $n = 0$ by setting $f_0 = 0$.

Recall that a nonnegative sequence $(f_n)_{n \geq 1}$ is log-concave with no internal zeros if $f_n^2 \geq f_{n-1}f_{n+1}$ whenever the three terms are defined, and its positive support is an interval of integers. We work throughout with formal power series, so no growth condition on (f_n) is needed.

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Theorem 1.1 (Cubic multiple-Pochhammer theorem). *Let $(f_n)_{n \geq 1}$ be a nonnegative log-concave sequence with no internal zeros, and put*

$$(1.2) \quad F_f(\mu; x) = \sum_{n \geq 1} f_n \frac{(\mu)_{3n}}{(3n-1)!} x^n.$$

For every $\mu, \alpha, \beta \geq 0$, the formal power series

$$(1.3) \quad \Delta_f(\mu; \alpha, \beta; x) = F_f(\mu + \alpha; x)F_f(\mu + \beta; x) - F_f(\mu; x)F_f(\mu + \alpha + \beta; x)$$

has nonnegative coefficients.

The proof has three components. The coefficient of degree m is rewritten as a beta-binomial expectation (Section 2). Two elementary monotonicity lemmas (Section 3) then supply the tools: beta laws with a fixed sum of parameters are ordered by their imbalance against symmetric functions increasing toward the midpoint, and a one-sign-change weighting principle carries positivity through nondecreasing weights. The weighted residue-class binomial kernel is shown to have the required monotonicity (Section 4); here complementary residue blocks have a single sign change, and the unweighted sum carries a Bernstein-basis positivity certificate. This kernel theorem does not depend on the beta-binomial setup. The components are combined in Section 5.

2. BETA-BINOMIAL REPRESENTATION

This section rewrites the coefficient convolution underlying the Turánian (1.3) as a beta-binomial expectation.

For $m \geq 2$, put

$$(2.1) \quad w_r = f_r f_{m-r}, \quad 1 \leq r \leq m-1.$$

These weights are symmetric. Log-concavity of (f_n) implies

$$w_{r+1} = f_{r+1} f_{m-r-1} \geq f_r f_{m-r} = w_r \quad \left(1 \leq r < \left\lfloor \frac{m}{2} \right\rfloor\right),$$

the standard cross-product consequence $f_{a+1} f_b \geq f_a f_{b+1}$ for $a < b$ of log-concavity with no internal zeros. With the symmetry $w_r = w_{m-r}$, the weights (2.1) satisfy the hypotheses of Theorem 4.1.

For $u, v \geq 0$, the degree- m coefficient of the product $F_f(u; x)F_f(v; x)$ is

$$(2.2) \quad C_{m,f}(u, v) = \sum_{r=1}^{m-1} w_r \frac{(u)_{3r} (v)_{3(m-r)}}{(3r-1)! [3(m-r)-1]!},$$

a finite sum, with $C_{m,f}(0, v) = 0$ since $(0)_{3r} = 0$. The beta-binomial representation below is used for $u, v > 0$. Let $N = 3m$ and let J have the beta-binomial distribution

$$\mathbb{P}(J = j) = \binom{N}{j} \frac{(u)_j (v)_{N-j}}{(u+v)_N}, \quad 0 \leq j \leq N.$$

For $1 \leq j \leq N-1$,

$$\frac{(u)_j (v)_{N-j}}{(j-1)! (N-j-1)!} = \frac{(u+v)_N}{N!} j(N-j) \mathbb{P}(J = j).$$

Defining a weight on all integers $0 \leq j \leq N$ by

$$\tilde{w}_j = \begin{cases} w_{j/3}, & 0 < j < N \text{ and } 3 \mid j, \\ 0, & \text{otherwise,} \end{cases}$$

we obtain

$$(2.3) \quad C_{m,f}(u, v) = \frac{(u+v)_N}{N!} \mathbb{E}[J(N-J) \tilde{w}_J].$$

The factor $J(N-J)$ vanishes at $j = 0$ and $j = N$, so the endpoint terms do not contribute.

Use the standard mixture representation $P \sim \text{Beta}(u, v)$ and $J \mid P \sim \text{Bin}(N, P)$. The identity

$$j(N-j) \binom{N}{j} = N(N-1) \binom{N-2}{j-1}$$

then transforms (2.3) into

$$(2.4) \quad C_{m,f}(u, v) = \frac{(u+v)^N}{N!} N(N-1) \mathbb{E} G_{m,w}(P),$$

where the kernel

$$(2.5) \quad G_{m,w}(p) = \sum_{r=1}^{m-1} w_r \binom{3m-2}{3r-1} p^{3r} (1-p)^{3(m-r)}, \quad 0 \leq p \leq 1,$$

is defined for arbitrary symmetric nonnegative weights $w_r = w_{m-r}$ and is symmetric under $p \mapsto 1-p$. The congruence $j \equiv 0 \pmod{3}$ becomes $j-1 \equiv 2 \pmod{3}$, which explains the binomial coefficient $\binom{3m-2}{3r-1}$.

By (2.4), Theorem 1.1 reduces to the monotonicity of $G_{m,w}$ on $[0, 1/2]$ (Section 4), via the imbalance ordering of Section 3.

3. TWO MONOTONICITY LEMMAS

This section records two elementary monotonicity facts: a weighting principle for the paired residue blocks of Section 4, of the kind used in the multiplicity-two analysis of [7], and an ordering of beta laws by imbalance used in Section 5, a fixed-sum likelihood-ratio comparison for folded beta laws [3].

Lemma 3.1 (One-sign-change weighting). *Let $0 \leq w_1 \leq w_2 \leq \dots \leq w_L$ and suppose the real sequence $A_1, \dots, A_L$ has no more than one sign change, necessarily from nonpositive to nonnegative. If*

$$\sum_{k=1}^L A_k \geq 0,$$

then

$$\sum_{k=1}^L w_k A_k \geq 0.$$

Proof. The claim is immediate if all A_k are nonnegative. Otherwise, let q be the largest index for which $A_q < 0$. Put $c = w_q$. For $k \leq q$ we have $w_k \leq c$ and $A_k \leq 0$, while for $k > q$ we have $w_k \geq c$ and $A_k \geq 0$. Therefore

$$\sum_{k=1}^L w_k A_k \geq c \sum_{k=1}^L A_k \geq 0.$$

□

Lemma 3.2 (Symmetric beta imbalance ordering). *Fix $s > 0$. For $0 \leq d < s/2$, let $P_d \sim \text{Beta}(s/2 + d, s/2 - d)$. If $H : [0, 1] \rightarrow \mathbb{R}$ is symmetric, $H(p) = H(1-p)$, and nondecreasing on $[0, 1/2]$, then*

$$(3.1) \quad d_1 \leq d_2 \implies \mathbb{E}H(P_{d_1}) \geq \mathbb{E}H(P_{d_2}).$$

If H is strictly increasing on $(0, 1/2)$ and $d_1 < d_2$, then the inequality is strict.

Proof. The case $d_1 = d_2$ is immediate, so assume $d_1 < d_2$. Put $Q_d = P_d(1 - P_d) \in (0, 1/4)$. For

$$p_{\pm}(q) = \frac{1 \pm \sqrt{1-4q}}{2}, \quad \ell(q) = \log \frac{p_+(q)}{p_-(q)},$$

the density of Q_d is

$$g_d(q) = \frac{2}{B(s/2 + d, s/2 - d)} q^{s/2-1} (1-4q)^{-1/2} \cosh(d\ell(q)).$$

The densities g_{d_1} and g_{d_2} share the factor $q^{s/2-1}(1-4q)^{-1/2}$, so

$$\frac{g_{d_2}(q)}{g_{d_1}(q)} = \frac{B(s/2 + d_1, s/2 - d_1)}{B(s/2 + d_2, s/2 - d_2)} \frac{\cosh(d_2 \ell(q))}{\cosh(d_1 \ell(q))}.$$

The function ℓ is strictly decreasing, since $p_+(q)/p_-(q)$ decreases to 1 as $q \uparrow 1/4$. For $d_2 > d_1$, $\cosh(d_2 \ell)/\cosh(d_1 \ell)$ is strictly increasing in $\ell > 0$, because $d \mapsto d \tanh(d\ell)$ is strictly increasing; hence g_{d_2}/g_{d_1} is strictly decreasing in q . Thus Q_{d_2} is smaller than Q_{d_1} in the likelihood-ratio order, and therefore in the usual stochastic order [11]. Since both densities integrate to one and their ratio is strictly monotone, it crosses 1 exactly once; hence $g_{d_1} - g_{d_2}$ has a single sign change, negative near 0 and positive near $1/4$, so for every nondecreasing ϕ ,

$$\mathbb{E}\phi(Q_{d_1}) - \mathbb{E}\phi(Q_{d_2}) = \int_0^{1/4} (\phi(q) - \phi(q^*)) (g_{d_1}(q) - g_{d_2}(q)) dq \geq 0,$$

where q^* is the crossing point and the integrand is nonnegative throughout. Finally, symmetry of H gives $H(p) = \tilde{H}(p(1-p))$, with $\tilde{H}$ nondecreasing on $[0, 1/4]$ precisely when H is nondecreasing on $[0, 1/2]$; taking $\phi = \tilde{H}$ proves (3.1). If H is strictly increasing and $d_1 < d_2$, the likelihood ratio is nonconstant, the crossing is nondegenerate, and the inequality is strict. $\square$

4. THE CUBIC RESIDUE KERNEL

For symmetric nonnegative weights $w_r = w_{m-r}$, the kernel of (2.5) is

$$G_{m,w}(p) = \sum_{r=1}^{m-1} w_r \binom{3m-2}{3r-1} p^{3r} (1-p)^{3(m-r)}, \quad G_{m,w}(p) = G_{m,w}(1-p).$$

This section proves that $G_{m,w}$ is monotone on $[0, 1/2]$ whenever the weights increase toward the center.

Theorem 4.1 (Weighted cubic residue monotonicity). *Suppose*

$$(4.1) \quad 0 \leq w_1 \leq w_2 \leq \cdots \leq w_{\lfloor m/2 \rfloor}, \quad w_r = w_{m-r}.$$

Then $G_{m,w}$ is nondecreasing on $[0, 1/2]$. If the weights are not all zero, then $G_{m,w}$ is strictly increasing on $[0, 1/2]$.

4.1. The constant-weight kernel. Let $w_r \equiv 1$, and write $G_m = G_{m,1}$. Set $n = 3m - 2$ and $t = p/(1-p)$. For $0 \leq p \leq 1/2$ one has $0 \leq t \leq 1$, and

$$G_m\left(\frac{t}{1+t}\right) = \frac{1}{(1+t)^{3m}} \sum_{r=1}^{m-1} \binom{n}{3r-1} t^{3r}.$$

Differentiation gives

$$(4.2) \quad \frac{d}{dt} G_m\left(\frac{t}{1+t}\right) = \frac{3J_m(t)}{(1+t)^{3m+1}},$$

where

$$(4.3) \quad J_m(t) = \sum_{r=1}^{m-1} \binom{n}{3r-1} t^{3r-1} (r - (m-r)t).$$

Lemma 4.2 (Bernstein certificate). *For every $m \geq 2$,*

$$J_m(t) > 0 \quad (0 < t < 1).$$

Consequently, G_m is strictly increasing on $[0, 1/2]$.

Proof. Writing $k = 3r - 1$ in (4.3) gives

$$J_m(t) = \frac{1}{3} \sum_{\substack{0 \leq k \leq n \\ k \equiv 2(3)}} \binom{n}{k} t^k ((k+1) - (n-k+1)t).$$

Introduce the projective Bernstein transform

$$(4.4) \quad \mathcal{P}_m(x) = (1+x)^{n+1} J_m\left(\frac{x}{1+x}\right).$$

Then

$$(4.5) \quad \mathcal{P}_m(x) = \frac{1}{3} \sum_{\substack{0 \leq k \leq j \\ k \equiv 2(3)}} \binom{n}{k} x^k (1+x)^{n-k} ((k+1) + (2k-n)x).$$

For $0 \leq j \leq n+1$, coefficient extraction from (4.5) gives

$$(4.6) \quad \begin{aligned} [x^j] \mathcal{P}_m(x) &= \frac{1}{3} \sum_{\substack{0 \leq k \leq j \\ k \equiv 2(3)}} \binom{n}{k} \left[(k+1) \binom{n-k}{j-k} + (2k-n) \binom{n-k}{j-k-1} \right] \\ &= \frac{\binom{n+1}{j}}{3(n+1)} S_{n,j}, \end{aligned}$$

where

$$(4.7) \quad S_{n,j} = \sum_{\substack{0 \leq k \leq j \\ k \equiv 2(3)}} \binom{j}{k} (n-k+1)(2k+1-j).$$

The second equality in (4.6) follows from the elementary identities

$$\binom{n}{k} \binom{n-k}{j-k} = \binom{n}{j} \binom{j}{k}, \quad \binom{n}{k} \binom{n-k}{j-k-1} = \binom{n+1}{j} \binom{j}{k} \frac{j-k}{n+1}.$$

The transform is Bernstein-theoretic: writing $J_m(t) = \sum_j b_j \binom{n+1}{j} t^j (1-t)^{n+1-j}$ in the degree- $(n+1)$ Bernstein basis, one has $[x^j] \mathcal{P}_m(x) = b_j \binom{n+1}{j}$, so nonnegativity of the Bernstein coefficients b_j certifies $J_m(t) \geq 0$ on $[0, 1]$ [10]. We claim that every $S_{n,j}$ is nonnegative.

Put $d = n+1-j$. A third-root-of-unity filter evaluates the sum over the class $k \equiv 2 \pmod{3}$; its trigonometric terms are periodic modulo 6 in j , so the closed form splits into six residue classes,

$$(4.8) \quad 3S_{n,j} = \begin{cases} d(2^j - 3j - 1) - 3j(j-1), & j \equiv 0 \pmod{6}, \\ d(2^j - 2) - 3j(j-1), & j \equiv 1 \pmod{6}, \\ d(2^j + 3j - 1), & j \equiv 2 \pmod{6}, \\ d(2^j + 3j + 1) + 3j(j-1), & j \equiv 3 \pmod{6}, \\ d(2^j + 2) + 3j(j-1), & j \equiv 4 \pmod{6}, \\ d(2^j - 3j + 1), & j \equiv 5 \pmod{6}. \end{cases}$$

In detail, for

$$R_a(j) = \sum_{k \equiv a(3)} \binom{j}{k}$$

and $\omega = e^{2\pi i/3}$,

$$R_a(j) = \frac{1}{3} \sum_{\ell=0}^2 \omega^{-a\ell} (1 + \omega^\ell)^j = \frac{1}{3} \left(2^j + 2 \cos \frac{(j-2a)\pi}{3} \right),$$

where $2 \cos(x\pi/3)$ runs through $2, 1, -1, -2, -1, 1$ as $x \equiv 0, 1, 2, 3, 4, 5 \pmod{6}$. The first two moments over the residue class are

$$\sum_{k \equiv 2(3)} k \binom{j}{k} = j R_1(j-1), \quad \sum_{k \equiv 2(3)} k^2 \binom{j}{k} = j(j-1) R_0(j-2) + j R_1(j-1).$$

Writing $d = n + 1 - j$ and expanding the summand in (4.7) gives, for $j \geq 2$,

$$(4.9) \quad \begin{aligned} S_{n,j} &= d(2jR_1(j-1) + (1-j)R_2(j)) \\ &\quad + j(j-1)(-2R_0(j-2) + 3R_1(j-1) - R_2(j)), \end{aligned}$$

with $j = 0, 1$ following directly from (4.7). Substituting the values of $R_2(j)$, $R_1(j-1)$, and $R_0(j-2)$ by residue $j \bmod 6$ yields (4.8).

It remains only to inspect the six cases. The cases $j \equiv 2, 3, 4 \pmod{6}$ are immediate. If $j \equiv 5 \pmod{6}$, then either $d = 0$, or $j \geq 5$ and $2^j - 3j + 1 > 0$. If $j \equiv 1 \pmod{6}$, then $d \geq 1$; the case $j = 1$ gives zero, while for $j \geq 7$,

$$(4.10) \quad 2^j - 2 \geq 3j(j-1).$$

Finally, $n + 1 = 3m - 1$ is congruent to 2 or 5 modulo 6. Thus, when $j \equiv 0 \pmod{6}$ and $j > 0$, one has $d \geq 2$, and for $j \geq 6$,

$$(4.11) \quad 2(2^j - 3j - 1) \geq 3j(j-1).$$

The inequalities (4.10) and (4.11) hold with equality at $j = 7$ and $j = 6$, respectively. Advancing j by one increases the left-minus-right difference by $2^j - 6j$ in (4.10) and by $2^{j+1} - 6j - 6$ in (4.11), both positive for $j \geq 6$; the inequalities therefore persist for all larger j . Likewise $2^j - 3j + 1 > 0$ at $j = 5$, with increment $2^j - 3 > 0$ thereafter. Hence $S_{n,j} \geq 0$ in every case.

By (4.6), all coefficients of $\mathcal{P}_m$ are nonnegative. The coefficient at x^2 is positive, since $d = n - 1 > 0$ and $3S_{n,2} = 9d$. Therefore $\mathcal{P}_m(x) > 0$ for $x > 0$. The substitution $t = x/(1+x)$ in (4.4) proves $J_m(t) > 0$ for $0 < t < 1$, and (4.2) completes the proof. $\square$

4.2. Insertion of log-concave weights. Arbitrary weights satisfying (4.1) preserve positivity of the derivative.

For $0 < t < 1$, differentiation of (2.5) gives

$$(4.12) \quad \frac{d}{dt} G_{m,w} \left(\frac{t}{1+t} \right) = \frac{3J_{m,w}(t)}{(1+t)^{3m+1}},$$

where

$$(4.13) \quad J_{m,w}(t) = \sum_{r=1}^{m-1} w_r \binom{3m-2}{3r-1} t^{3r-1} (r - (m-r)t).$$

Pair the terms indexed by r and $m-r$. For $1 \leq r < m/2$, let

$$(4.14) \quad \begin{aligned} B_{m,r}(t) &= \binom{3m-2}{3r-1} \left[t^{3r-1} (r - (m-r)t) \right. \\ &\quad \left. + t^{3(m-r)-1} ((m-r) - rt) \right]. \end{aligned}$$

If m is even, the central block is

$$(4.15) \quad B_{m,m/2}(t) = \binom{3m-2}{3m/2-1} \frac{m}{2} t^{3m/2-1} (1-t) > 0.$$

Then

$$J_{m,w}(t) = \sum_{r=1}^{\lfloor m/2 \rfloor} w_r B_{m,r}(t),$$

with the convention (4.15) at the center.

Lemma 4.3 (Single sign change of the paired blocks). *For each fixed $m \geq 2$ and $0 < t < 1$, the sequence*

$$B_{m,1}(t), B_{m,2}(t), \dots, B_{m,\lfloor m/2 \rfloor}(t)$$

has no more than one sign change, from negative to positive.

Proof. For $r < m/2$, the sign of $B_{m,r}(t)$ is the sign of

$$H_{m,r}(t) = r - (m - r)t + t^{3(m-2r)}((m - r) - rt).$$

Put $d = m - 2r > 0$ and $t = e^{-x}$, $x > 0$. Direct rearrangement gives

$$\begin{aligned} H_{m,r}(t) &= \frac{1}{2}(1+t)(1+t^{3d}) \left[m \frac{1-t}{1+t} - d \frac{1-t^{3d}}{1+t^{3d}} \right] \\ (4.16) \quad &= \frac{1}{2}(1+t)(1+t^{3d}) \left[m \tanh \frac{x}{2} - d \tanh \frac{3dx}{2} \right]. \end{aligned}$$

The function $d \mapsto d \tanh(3dx/2)$ is strictly increasing for $d > 0$. Therefore, for $2 \leq r < m/2$, if $B_{m,r}(t) \leq 0$, then the same bracket in (4.16) is strictly negative after replacing d by $d + 2$, which corresponds to replacing r by $r - 1$. Thus

$$B_{m,r}(t) \leq 0 \implies B_{m,r-1}(t) < 0 \quad (2 \leq r < m/2).$$

The final block is always positive. For even m this is (4.15). For odd m , the final block has $d = 1$, and its bracket in (4.16) is $m \tanh(x/2) - \tanh(3x/2) > 0$, because $m \geq 3$ and $\tanh(3y) < 3 \tanh y$ for $y > 0$. Hence the blocks have the asserted sign pattern. $\square$

Proof of Theorem 4.1. By Lemma 4.2,

$$\sum_{r=1}^{\lfloor m/2 \rfloor} B_{m,r}(t) = J_m(t) > 0 \quad (0 < t < 1).$$

The block sequence has one sign change by Lemma 4.3. Applying Lemma 3.1 with the nondecreasing weights in (4.1) gives

$$J_{m,w}(t) = \sum_{r=1}^{\lfloor m/2 \rfloor} w_r B_{m,r}(t) \geq 0.$$

Equation (4.12) proves that $G_{m,w}$ is nondecreasing on $[0, 1/2]$.

Suppose now that the weights are not all zero, and put $L = \lfloor m/2 \rfloor$. Then $w_L > 0$, and $B_{m,L}(t) > 0$ by Lemma 4.3. If every block is nonnegative, then $J_{m,w}(t) \geq w_L B_{m,L}(t) > 0$. Otherwise let q be the last index with $B_{m,q}(t) < 0$, and put $c = w_q$. The decomposition

$$\sum_{r=1}^L w_r B_{m,r} = c \sum_{r=1}^L B_{m,r} + \sum_{r \leq q} (w_r - c) B_{m,r} + \sum_{r > q} (w_r - c) B_{m,r}$$

displays the weighted sum as three nonnegative terms: the first because $\sum_r B_{m,r} = J_m(t) > 0$ and $c \geq 0$; the second because $w_r \leq c$ and $B_{m,r} \leq 0$ for $r \leq q$; the third because $w_r \geq c$ and $B_{m,r} \geq 0$ for $r > q$. If $c > 0$, the first term is strictly positive; if $c = 0$, the third term is strictly positive, because $q < L$ and it contains $(w_L - c) B_{m,L}(t) = w_L B_{m,L}(t) > 0$. Hence $J_{m,w}(t) > 0$ for $0 < t < 1$, and $G_{m,w}$ is strictly increasing on $[0, 1/2]$. $\square$

5. PROOF OF THE THEOREM

The representation (2.4) and the two monotonicity facts of Sections 3 and 4 combine to show that the coefficient convolution is Schur-concave in its two parameters [8] — equivalently, nonincreasing under increasing imbalance at fixed parameter sum — which gives Theorem 1.1.

Proposition 5.1 (Schur-concavity of the coefficient convolution). *For each $m \geq 2$ and each log-concave sequence (f_n) as in Theorem 1.1, the symmetric function $(u, v) \mapsto C_{m,f}(u, v)$ is Schur-concave on $(0, \infty)^2$. Equivalently, at fixed $s = u + v$,*

$$d \longmapsto C_{m,f} \left(\frac{s}{2} + d, \frac{s}{2} - d \right)$$

is nonincreasing for $0 \leq d < s/2$. If at least one product $f_r f_{m-r}$ is positive, then this function is strictly decreasing.

Proof. The prefactor in (2.4) depends only on $s = u + v$. By Theorem 4.1, $G_{m,w}$ is symmetric and nondecreasing on $[0, 1/2]$. The conclusion follows immediately from Lemma 3.2. If at least one weight $w_r = f_r f_{m-r}$ is positive, then Theorem 4.1 makes $G_{m,w}$ strictly increasing, and the strict part of Lemma 3.2 gives strict decrease in d . $\square$

Proof of Theorem 1.1. The coefficients of degrees 0 and 1 in (1.3) vanish. For $m \geq 2$, direct multiplication gives

$$(5.1) \quad [x^m] \Delta_f(\mu; \alpha, \beta; x) = C_{m,f}(\mu + \alpha, \mu + \beta) - C_{m,f}(\mu, \mu + \alpha + \beta).$$

Assume first that $\mu > 0$ and $\alpha, \beta > 0$. The two parameter pairs in (5.1) have the same sum

$$s = 2\mu + \alpha + \beta.$$

Their imbalances are

$$d_1 = \frac{|\alpha - \beta|}{2}, \quad d_2 = \frac{\alpha + \beta}{2},$$

and $d_1 \leq d_2$. Proposition 5.1 therefore gives

$$C_{m,f}(\mu + \alpha, \mu + \beta) \geq C_{m,f}(\mu, \mu + \alpha + \beta).$$

This proves nonnegativity of every coefficient. If $\alpha = 0$ or $\beta = 0$, the Turánian is identically zero. The boundary case $\mu = 0$ follows either by continuity in μ or directly from $F_f(0; x) = 0$ and the nonnegative coefficients of $F_f(\alpha; x)F_f(\beta; x)$. $\square$

Corollary 5.2 (Strict coefficient classification). *Under the hypotheses of Theorem 1.1, let $\mu \geq 0$, $\alpha, \beta > 0$, and $m \geq 2$. Then*

$$[x^m] \Delta_f(\mu; \alpha, \beta; x) > 0$$

if and only if there exists $1 \leq r \leq m - 1$ with $f_r f_{m-r} > 0$.

Proof. If every product $f_r f_{m-r}$ vanishes, then both terms of (5.1) vanish. Conversely, suppose one such product is positive. If $\mu > 0$, the two imbalances in the proof of Theorem 1.1 satisfy

$$\frac{|\alpha - \beta|}{2} < \frac{\alpha + \beta}{2},$$

so the strict part of Proposition 5.1 gives strict inequality. If $\mu = 0$, then $F_f(0; x) = 0$, so the coefficient equals $C_{m,f}(\alpha, \beta)$, a sum of nonnegative terms with at least one strictly positive. $\square$

Corollary 5.3 (Ordinary log-concavity). *Suppose (f_n) is not identically zero, and let R be the radius of convergence of $\sum_{n \geq 1} f_n x^n$. For every $0 < x < R$, the function $\mu \mapsto F_f(\mu; x)$ is finite, continuous, and positive on $(0, \infty)$, and $\mu \mapsto \log F_f(\mu; x)$ is concave there.*

Proof. For each fixed $\mu > 0$, Stirling's formula gives

$$\frac{(\mu)_{3n}}{(3n-1)!} = \frac{\Gamma(3n + \mu)}{\Gamma(\mu)\Gamma(3n)} \sim \frac{(3n)^\mu}{\Gamma(\mu)} \quad (n \rightarrow \infty).$$

Hence $F_f(\mu; \cdot)$ has the same radius of convergence R as $\sum_{n \geq 1} f_n x^n$. On every compact set of positive μ and for $0 < x < R$, the series converges locally uniformly in μ , since its extra factor grows at most polynomially in n ; thus $\mu \mapsto F_f(\mu; x)$ is continuous. Since (f_n) is not identically zero, its nonnegative coefficients give $F_f(\mu; x) > 0$ for $\mu, x > 0$. By Theorem 1.1,

$$F_f(\mu + \alpha; x)F_f(\mu + \beta; x) \geq F_f(\mu; x)F_f(\mu + \alpha + \beta; x)$$

for all nonnegative shifts. Taking $\alpha = \beta = h > 0$ gives

$$F_f(\mu + h; x)^2 \geq F_f(\mu; x)F_f(\mu + 2h; x),$$

so $\mu \mapsto \log F_f(\mu; x)$ is midpoint concave, and by continuity concave, on $(0, \infty)$. $\square$

6. CONCLUDING REMARKS

Remark 6.1 (Why the cubic case closes). The proof hinges on the special feature of the residue class modulo 3. After the beta-binomial reduction, the entire conjecture is controlled by the monotonicity of

$$p(1-p)\mathbb{P}\{\text{Bin}(3m-2, p) \equiv 2 \pmod{3}\}.$$

The projective transform (4.4) converts its derivative into a polynomial whose Bernstein coefficients are nonnegative. This positivity is not termwise obvious in the monomial basis; it appears only after the degree lift and the mod-6 root-of-unity evaluation.

Remark 6.2 (The role of log-concavity). The log-concavity assumption is used exactly once: it makes the symmetric weights $w_r = f_r f_{m-r}$ nondecreasing as r approaches $m/2$. The paired derivative blocks have a single sign change, so Lemma 3.1 transfers the constant-weight positivity certificate to the full weighted kernel.

Remark 6.3 (Higher multiplicities). For a general integer r , the same beta-binomial reduction produces, in the constant-weight case,

$$p(1-p)\mathbb{P}\{\text{Bin}(rm-2, p) \equiv r-1 \pmod{r}\},$$

and in general its analogue weighted by the products $f_k f_{m-k}$. The cubic proof suggests testing whether a suitable projective degree lift has a positive coefficient array. The known failure at $r = 4$ shows that such positivity cannot persist without an additional restriction; locating the first coefficient obstruction in the lifted array may explain that threshold.

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